# CO4 - AT3

## Reverse Engineering Task.

Name :- Karthik. L
Reg NO :- 192573008

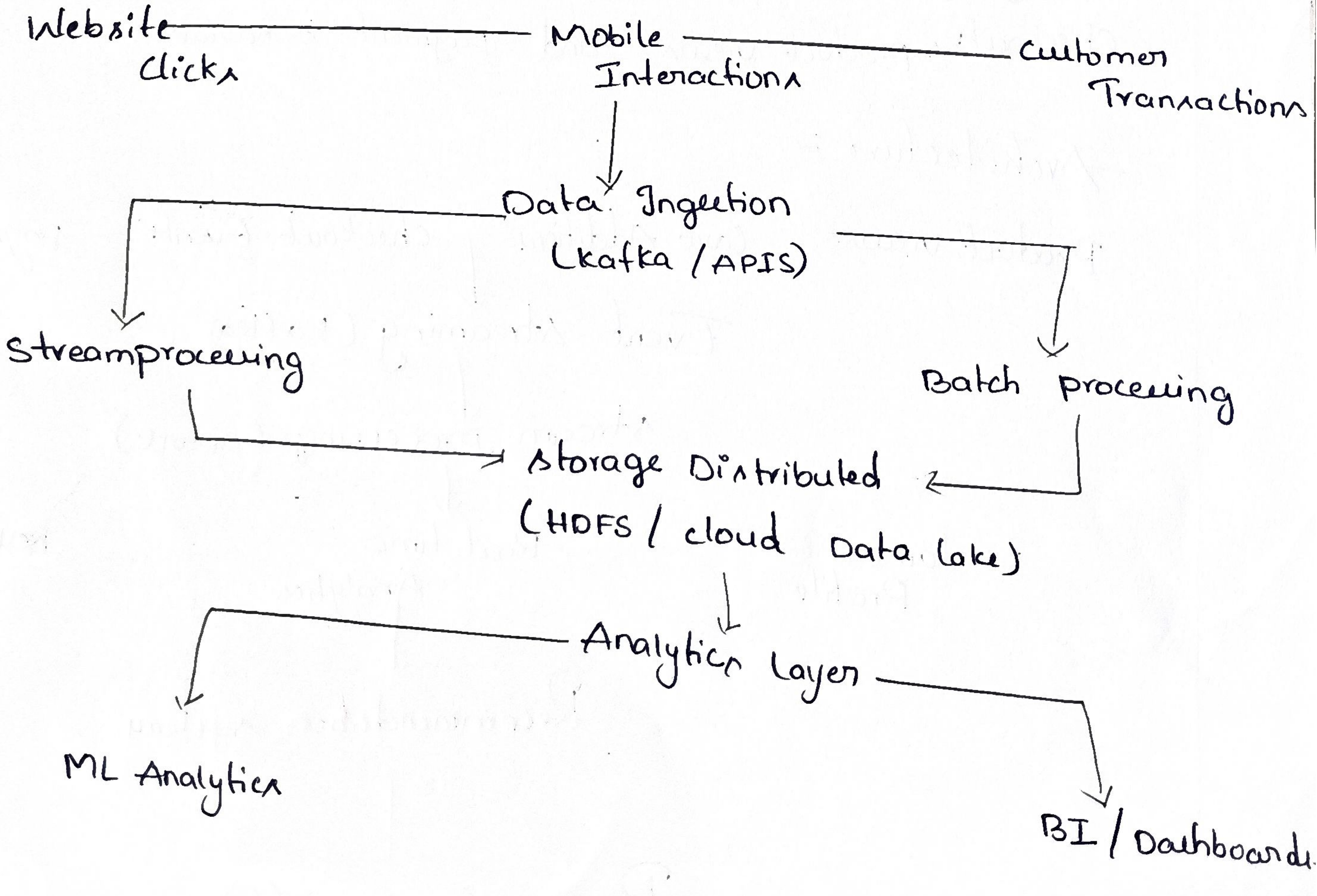
Course-code :- CSA1506
Course - name :- cloud computing

---

1. Big Data Platform for website clicks and Customer Transactions

A platform collects billions of websites clicks, mobile interactions, and customer transactions. A probable architecture would use distributed storage, stream processing.

Architecture :-



Distributed storage :-

A storage distributed file system such as HDFS (or) a cloud Data lake can divide large datasets into blocks and store them across multiple machines.

Data Organization :-

structured data such as customer transactions can be stored in relational databases (or) structured tables.

Semi - structured data :- such as click logs and mobile events can be stored in formats such as JSON (or) Parquet in the datalink

Scalability and partitioning :-

Data can be partitioned by date, region, customer, (or) even type.

Dashboard connections :- Sales, customer behavior, website activity, Business KPIs, Realtime reports.

2. E- Commerce platform :-

The E- Commerce platform tracks cart addition, abandoned Checkouts, product views, and payment behavior across regions.

Architecture :-

Product views — Cart Additions — checkout Events — Payment Events

↓

Event streaming (kafka)

↓

stream processing (spark)

↓

customer Profile ← Real time Analytics → ML models

↓

Recommendation systems

↓

Personalized offers.

Customer profiling :

Customer information can be maintained in a NoSQL database for fast access, while structured order and payment info- can be maintained in a relational database.

Synchronization :

1. Session data, click stream logs, Transaction records.

Recommendation workflow :

Customer Activity Data collection customer profile ML model

Recommendation products Personalized websites.

3. Bigdata Healthcare system :-

The healthcare system processes patient histories, diagnostic images, wearable sensors streams, prescription records. Hybrid Architecture is suitable for managing structured, semi-structured, and unstructured helthcare info.

Hybrid database architecture :-

structured info :- patient IDs, Prescription records, lab results

unstructured info :- diagnostic images, medical docs.

Streaming and ML :

-> Diseases prediction

-> Risk identification

-> Patient monitoring

-> Population health analysis.

Security Architecture :-

Patient records — Precription data — labdata — Diagnostic images — wearable sensors

↓

Data Integration

↓ Structured Storage

↓ Unstructured Data lake

↓

Distributed processing

↓

ML + Analytics

↓

Clinical Decision support.

↓

Security and privacy :-

→ Encryption

→ Authentication

→ Role-based access control

→ Audit logs

→ controlled data access

Analytics pipline :-

Healthcare data

↓

Integration

↓

Secure storage

↓

Distributed processing

↓

Analytics / ML

↓

Clinical Insights.

4. Smart City Platform

The smart city platform continously receives traffic sensor data, CCTV metadata, weather updates, and public transport information. A combination of edge computing, messaging, distributed storage, streaming analytics, and geospatial processing is suitable.

Architecture :-

Traffic sensors — CCTV Metadata — weather Data — Transport Data

↓

Edge Computing

↓

Distributed Messaging (kafka)

↓

Stream Processing

↓

Traffic Analytics ← weather Analytics → Transport Analytics

↓

Distributed storage

↓

Geospatial Analytics

↓

Live city Dashboard

## 5| Healthcare Analytics System

Edge Computing:-

Edge devices can process some sensors information near the location where it is generated.

Advantages:-

→ lower latency

→ Reduced network traffic

→ Faster response.

Geospatial analytics

→ Traffic congestion

→ Vehicle locations

→ Road conditions

→ Transport activity.

Databases:-

A time-series database can store continuously changing sensor measurements.

Batch + real-time analytics:-

Batch analytics can analyze historical traffic patterns for long-term urban planning.

Security:-

→ Authentication

→ Role-based access control

→ Encryption

→ Network security

→ Access monitoring.

Visualization:-

→ Current traffic conditions

→ Public transport status

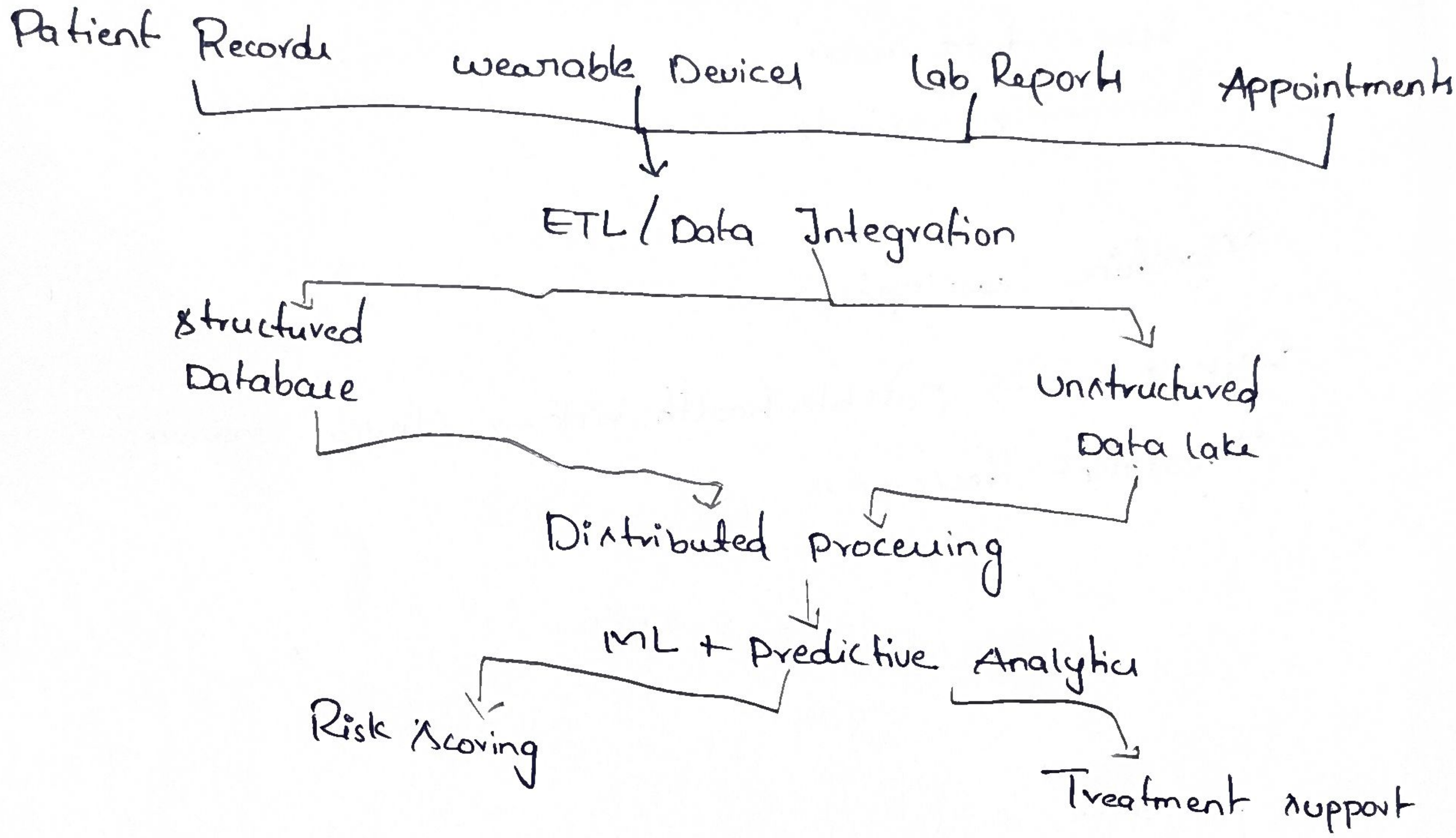
→ weather conditions

→ Congestion alerts.

5. Healthcare Analytics System

This healthcare analytics system combines patient records, wearable readings, laboratory reports, and appointment histories.

Architecture :-




Hybrid storage :-

→ Structured patient information can be stored in relational databases.

Interoperability :-

Common healthcare data standards and APIs can help different hospitals and clinics exchange information.

Distributed processing :-

→ Population health analysis.

→ Disease trends

→ Risk analysis

→ Healthcare planning.

Security and Compliance :

→ Data encryption.

→ Authentication

→ Authorization

→ Audit trails

→ Secure data access

→ Compliance controls.

Machine Learning :

Risk score → Possible health risk → Clinical review → Treatment decision.